

Topic: Union Operation

Q. Explain the Union operation in Relational Algebra. What is Union Compatibility? Provide an example to illustrate how the Union operator works.

Definition to Remember

The **Union Operation** ($\cup$) is a **binary set operation** in relational algebra that combines all tuples (rows) from two relations into a single result, automatically **eliminating duplicate rows**. The two relations must be **Union Compatible** (same number of columns and matching data types).

1. Union Operator ($\cup$)

- **Binary operation** — operates on two tables.
- **Syntax:** Relation_A $\cup$ Relation_B
- **Duplicate Elimination:** If a tuple exists in both A and B, it appears only **once** in the output.
- Equivalent to the `UNION` keyword in SQL.

2. Union Compatibility (Required Condition)

For Union (and also Intersection and Set Difference) to be valid, both relations **MUST be Union Compatible**:

Condition	Meaning
Same Degree	Both relations must have the same number of columns .
Same Domain	Corresponding columns must have matching data types (int with int, string with string, etc.).

3. Example

CRICKET_TEAM

Student_ID	Name
1	Alice
2	Bob
3	Charlie

FOOTBALL_TEAM

Student_ID	Name
3	Charlie
4	David
5	Eve

Query: Find all students who play Cricket OR Football or both.

`$CRICKET_TEAM  $\cup$  FOOTBALL_TEAM$`

Result: Alice, Bob, Charlie, David, Eve — **5 rows** (*Charlie appeared in both tables but appears only once — duplicate eliminated*).

★ **Must-Write Points (for 10 marks)**

1. Union ($\cup$) is a binary set operation that combines rows from two relations.
 2. Duplicate rows are automatically eliminated from the result.
 3. **Union Compatibility** is required: same number of columns AND matching data types.
 4. If $|A| = m$ rows and $|B| = n$ rows, $|A \cup B| \leq m + n$ (due to duplicate removal).
 5. Equivalent to SQL `UNION` (which also removes duplicates).
 6. Same Union Compatibility condition applies to Intersection and Set Difference.
 7. Union is used to merge data from similar tables (e.g., combining employee lists from two branches).
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⚡ **Quick Recall**

$\cup$ (Union) $\rightarrow$ Binary $\rightarrow$ Combines All Rows $\rightarrow$ Removes Duplicates $\rightarrow$ Union Compatibility (same degree + same domain) $\rightarrow$ SQL `UNION`